

Latest in CA/Browser Forum Tracker -- diff report

Generated: 2026-09-01T10:23:09.713758 | **Period:** since 2026-09-01 | **Leaves with news:** 9 of 15

Fresh developments: 21 (as_of within 14 days of the window) | **Superseded/conflicting:** 9 | **Background captured:** 64 (older material, first time in the knowledge base)

Reconfirmations in this period (no new information, not shown): 11.

Summary

Leaf	Fresh	Superseded	Conflicting	Background
CCADB, incident reports & distrust	6	0	0	17
Certificate Transparency	6	0	0	5
Root store policies	4	0	1	7
TLS BR ballots & versions	0	4	0	4
Validation Subcommittee & domain control	2	1	0	10
Certificate profiles & cryptographic requirements	2	0	0	3
Code Signing BR ballots & versions	0	2	0	3
S/MIME BR ballots & versions	1	1	0	1
Revocation mechanisms & timelines	0	0	0	14

Public trust programs

CCADB, incident reports & distrust (ccadb-incidents-distrust)

New developments (6)

CA incident reports on Bugzilla and their follow-through

- D-Trust incident report (bug 2029013) includes an ongoing action item with due date 2026-09-30 to reevaluate requirements and implement controls for CA/Browser Forum and Root Program compliance. ([D-Trust: Missing Pre-Signing Linting for TLS Issuance](#), as of 2026-09-01, confidence: medium, source: grok)
- *New specific incident and remediation timeline not previously recorded.*

CCADB policy and data conventions

- The CCADB Steering Committee proposed further policy updates (post v2.1) clarifying subordinate CA ownership disclosure, additional PKI policy document disclosures effective September 15, 2026, audit sampling methodology disclosure for periods beginning January 15, 2027, and Qualified Auditor involvement in incident reports. ([Further Improving the CCADB Policy and Incident Reporting Guidelines \(FEEDBACK REQUESTED\)](#), as of 2026-09-01, confidence: medium, source: grok)
- *New proposal (VERIFY status) not previously present in the knowledge base.*

The incident reporting regime

- When a CA suffers an incident it must file a report on Bugzilla that is detailed, candid, timely and transparent, demonstrating that the CA understands the root cause and is acting to resolve it. The quality of the report is itself part of what root programs judge. ([theregister.com — mozilla entrust](#), as of 2026-09-01, confidence: medium, source: ingest)
- *New general fact about the incident reporting regime's expectations, not previously recorded.*
- Mozilla's proposed updates to address root CA life cycles were taken to the dev-security-policy mailing list, which remains the venue where root program policy changes are argued before adoption. ([groups.google.com — qSvEhaHs1c](#), as of 2026-09-01, confidence: medium, source: ingest)
- *New fact confirming the ongoing role of dev-security-policy as the deliberation venue for root program policy changes.*

mozilla dev-security-policy discussion

- Mozilla dev-security-policy list discussions in 2026 covered MRSP v3.1 audit improvements, root lifecycle transitions, and responses to the May 2026 CA Survey. ([dev-security-policy@mozilla.org - Google Groups](#), as of 2026-09-01, confidence: medium, source: grok)
- *Adds the previously unrecorded detail of a May 2026 CA Survey being discussed on the list, beyond the general venue fact already known.*

root program distrust and removal actions

- Chrome Root Program Policy version 1.8 requires CA Owners with more than two self-signed root CA certificates in the Chrome Root Store to submit a consolidation plan before June 15, 2026, with a maximum of two self-signed roots per CA Owner effective September 15, 2027. ([Chrome Root Program Policy, Version 1.8](#), as of 2026-08-28, confidence: high, source: grok)
- *New Chrome Root Program policy detail on root consolidation requirements, not previously present.*

Newly captured background -- older material, first time in the knowledge base (17)

CA incident reports on Bugzilla and their follow-through

- Recent Bugzilla incident reports include NETLOCK non-compliance with CCADB Incident Reporting Guidelines on 'Related Incidents' section (bug opened 2026-01-29) and CFCA OCSP Responder certificate profile deviations identified April 2026. ([NETLOCK: Missing Related Incidents section in the bug report](#), as of 2026-01-29, confidence: high, source: grok)
- *New concrete incident-report examples not previously in the knowledge base.*

CCADB policy and data conventions

- On March 20, 2026, CCADB was updated to support a new editable JSON Array of all Full CRL URLs field for certificate records, new Automation Solution Attestation and Public Test Infrastructure URL fields for root records, and an Apple Root Program Policy Agreement field in Root Inclusion Request cases. ([CCADB Update: Support multiple full CRL URLs and other enhancements](#), as of 2026-03-20, confidence: high, source: grok)
- *New specific data-field/schema detail about the CCADB March 2026 update not previously recorded.*
- CCADB Policy version 2.1, effective March 20, 2026, requires CA policy documents to state adherence to the CCADB Policy effective September 15, 2026, and encourages focused single-use-case policy documents. ([Common CA Database Policy](#), as of 2026-03-20, confidence: high, source: grok)
- *Adds substantive new requirements of CCADB Policy v2.1 beyond the mere fact of its effective date already recorded.*
- CCADB Self-Assessment template version 1.8 became effective February 26, 2026. ([Common CA Database Self-Assessment](#), as of 2026-02-26, confidence: high, source: grok)
- *New fact about the CCADB self-assessment template version and effective date.*

Punitive distrust: the Entrust precedent

- Google was the first root program to drop Entrust, citing a pattern of concerning behaviours rather than any single incident. ([theregister.com — mozilla entrust](#), as of 2024-08-01, confidence: medium, source: ingest)
- *New fact about the sequencing of Entrust distrust actions across root programs.*
- Mozilla stopped trusting Entrust-issued certificates dated after 30 November 2024, while certificates issued before that date remained trusted. Distrust in this ecosystem is applied by issuance date rather than by immediate revocation. ([digicert.com — faq](#), as of 2024-11-30, confidence: medium, source: ingest)
- *New detail establishing the mechanism and effective date of Mozilla's Entrust distrust.*
- Entrust's compliance record included multiple Bugzilla-reported issues, among them TLS certificates issued with ECC P-384 keys hashed using SHA-256, contrary to Mozilla policy requiring SHA-384. ([theregister.com — mozilla entrust](#), as of 2024-08-01, confidence: medium, source: ingest)
- *New specific detail on the type of compliance issues that fed into the Entrust distrust decision.*

Scheduled root retirement, not punitive

- On 15 April 2026 Mozilla and Google Chrome removed the DigiCert G1 generation of roots from their trust stores, and TLS certificates chaining solely to those roots lost trust in Firefox and Chrome on that date. ([docs.digicert.com — review the g1 root removal advisory](#), as of 2026-04-15, confidence: medium, source: ingest)
- *First record of the DigiCert G1 root removal event in this leaf's knowledge base.*
- Three roots were affected: DigiCert Global Root CA, DigiCert Assured ID Root CA and DigiCert High Assurance EV Root CA. Affected certificates had to move to DigiCert Global Root G2 or G3. ([docs.digicert.com — review the g1 root removal advisory](#), as of 2026-04-15, confidence: medium, source: ingest)
- *New specific detail on which roots were removed and the migration target.*
- This removal is an age-based retirement supporting modern cryptographic standards rather than a response to CA misconduct, and it is the practical expression of the root programs' move against indefinitely trusted roots. ([docs.digicert.com — review the g1 root removal advisory](#), as of 2026-04-15, confidence: medium, source: ingest)
- *New characterization distinguishing scheduled retirement from punitive distrust actions, useful context for the leaf's scope.*

The incident reporting regime

- CCADB Policy version 2.1 took effect on 20 March 2026 and governs how CA operators record and maintain their data in the Common CA Database. ([github.com — policy](#), as of 2026-03-20, confidence: medium, source: ingest)
- *New fact establishing the current CCADB policy version and effective date.*

mozilla dev-security-policy discussion

- Mozilla Root Store Policy version 3.1 was published June 29, 2026 and effective July 1, 2026, introducing requirements for Detailed Controls Reports (DCRs) beginning with audit periods starting on or after July 1, 2027 for TLS website authentication roots. ([Improving Transparency and Assurance in the Web PKI: Mozilla Root Store Policy v3.1](#), as of 2026-06-29, confidence: high, source: grok)
- *New fact establishing MRSP v3.1's publication, effective date and DCR requirement, not previously in the base.*

root program distrust and removal actions

- Mozilla Root Store Policy v3.1 effective July 1, 2026 limits new root inclusion requests to root CA key pairs generated no more than five years prior to submission. ([Mozilla Root Store Policy](#), as of 2026-07-01, confidence: high, source: grok)

Conflicting -- needs review (1)

Chrome Root Program Policy

- Chrome Root Program Policy reached version 1.8 (last updated 2026-02-05), adding requirements to limit the number of root certificates per CA Owner, extend automation support to all existing CAs, require CP/CPS documents to explicitly state adherence to the policy, set expectations for industry engagement, recommend Subordinate CA lifecycle practices, encourage CT ecosystem support, and consolidate Applicant-only requirements. ([Chrome Root Program Policy, Version 1.8](#), as of 2026-02-05, confidence: high, source: grok)
- conflicts with: "[d0001:7] Chrome Root Program Policy version 1.6 requires dedicated hierarchies by June 2026"
- *A version 1.8 dated 2026-02-05 predates the June 2026 date attached to version 1.6, an inconsistent version/date ordering that cannot be reconciled without human review.*

Newly captured background -- older material, first time in the knowledge base (7)

CCADB

- CCADB Policy version 2.1 took effect on 20 March 2026. CA operators with certificates in Mozilla's root store must use the CCADB and comply with its policy, and Mozilla imposes further CCADB requirements beyond it. ([github.com — policy](#), as of 2026-03-20, confidence: medium, source: ingest)
- *First record of CCADB Policy versioning and its relation to Mozilla's root store requirements.*

Chrome Root Program

- Chrome Root Program Policy version 1.6 requires that certificate hierarchies included in Chrome's trust store be dedicated solely to TLS server authentication by June 2026. ([chromium.org — root ca policy](#), as of 2026-06-30, confidence: medium, source: ingest)
- *New Chrome Root Program version and dedicated-hierarchy deadline, distinct from Mozilla's own dedicated-hierarchy rule.*

Microsoft Trusted Root Program

- Microsoft Trusted Root Program Requirements (effective July 1, 2026) require newly minted Root CAs to be valid for a maximum of 10 years from the date of submission and mandate that participating Root CAs be limited in scope and dedicated to clearly defined trust purposes. ([Microsoft Root Program Requirements v1.1](#), as of 2026-07-01, confidence: high, source: grok)
- *First recorded Microsoft Trusted Root Program key-lifetime and dedicated-purpose requirements; no prior Microsoft program facts exist in this leaf.*

Mozilla Root Store Policy

- Mozilla Root Store Policy version 3.1 took effect on 1 July 2026, published under the theme of improving transparency and assurance in the Web PKI. ([blog.mozilla.org — improving transparency and assurance in the web pki mozilla root store policy v3 1](#), as of 2026-07-01, confidence: medium, source: ingest)
- *First record of MRSP version 3.1 and its effective date; nothing on Mozilla policy versions exists in the base.*
- MRSP 3.1 revises Section 3.3 to require that CP and CPS documentation be explicit, bounded, auditable and sufficiently detailed to describe certificate issuance activities, adding requirements for version control, accessibility and ongoing maintenance. ([blog.mozilla.org — improving transparency and assurance in the web pki mozilla root store policy v3 1](#), as of 2026-07-01, confidence: medium, source: ingest)
- *New detail on specific CP/CPS documentation requirements introduced by MRSP 3.1.*
- MRSP 3.1 also requires root CA key pairs to have been generated within the previous five years, aligns mass revocation planning with the Forum's Baseline Requirements, clarifies audit expectations for root inclusion

requests, and clarifies notification when CA ownership or operational control changes. ([blog.mozilla.org — improving transparency and assurance in the web pki mozilla root store policy v3 1](#), as of 2026-07-01, confidence: medium, source: ingest)

- *New key-lifetime limit and other MRSP 3.1 provisions not previously recorded.*
- Mozilla requires new root CA certificates to be dedicated to either TLS or S/MIME. CA operators with existing dual-purpose roots had to submit a transition plan by 15 April 2026 and must complete migration to separate roots by 31 December 2028. ([mozilla.org — policy](#), as of 2026-04-15, confidence: medium, source: ingest)
- *New dedicated-hierarchy requirement with specific transition and migration dates, distinct from the general MRSP 3.1 content.*

Revocation & transparency

Certificate Transparency (~~certificate-transparency~~)

New developments (6)

Chrome CT policies and SCT requirements

- Chrome Root Program Policy v1.8 requires CAs to log all TLS server authentication precertificates to at least one Usable/Qualified CT log effective June 15, 2026. ([Chrome Root Program Policy, Version 1.8](#), as of 2026-08-28, confidence: high, source: grok)
- *New requirement and effective date from Chrome Root Program Policy v1.8, not previously recorded.*

Log operator lifecycle (inclusion, freezing, retirement, temporal sharding)

- Chrome CT log policy documents define lifecycle states (Pending, Qualified, Usable, ReadOnly, Retired, Rejected) with temporal sharding requirements for log expiry ranges. ([CertificateTransparency/log_states.md](#), as of 2026-09-01, confidence: high, source: grok)
- *First detailed record of the formal log lifecycle state model, not previously present in the knowledge base.*

RFC 6962 to RFC 9162 and the static-CT / tiled log API

- Static CT API specification and tiled log implementations (Sunlight, Azul, TesseraCT) continue active development with temporal sharding for certificate expiry windows. ([C2SP/static-ct-api.md](#), as of 2026-09-01, confidence: high, source: grok)
- *New detail naming specific tiled log implementations and confirming temporal sharding practice, beyond the prior version-history note.*

The static-ct-api transition

- Chrome's Certificate Transparency log policy accepts logs conforming either to RFC 6962 or to the static-ct-api v1.1.0 C2SP specification. Both are admissible; neither has been retired. ([googlechrome.github.io — log policy](#), as of 2026-09-01, confidence: medium, source: ingest)
- *Establishes the current dual-format state of Chrome's CT log policy, not present in the empty baseline knowledge base.*
- The Static CT API reached v1.0.0-rc.1 alongside Sunlight v0.3.1, the reference implementation, before the v1.1.0 specification Chrome's policy now names. ([groups.google.com — 936IR3MEUDU](#), as of 2026-09-01, confidence: medium, source: ingest)
- *New historical/version detail about the static-ct-api specification's development lineage, not previously recorded.*

crt.sh and monitor ecosystem changes

- crt.sh maintains active GitHub repositories (e.g., ctsubmit, ctlint, crtsh-web) with updates as recent as September 1, 2026. ([crtsh repositories](#), as of 2026-09-01, confidence: high, source: grok)
- *New information about crt.sh tooling ecosystem activity not previously recorded.*

Newly captured background -- older material, first time in the knowledge base (5)

Chrome CT policies and SCT requirements

- Chrome will remove support for OCSP-delivered SCTs in Chrome 148 (releasing May 5, 2026). ([Change to Chrome's CT policy: removing support for OCSP-delivered SCTs](#), as of 2026-03-20, confidence: high, source: grok)
- *New Chrome CT policy change (removal of OCSP-delivered SCT support) not previously recorded.*
- Chrome removed the one-RFC6962-log requirement; as of April 15, 2026, SCTs exclusively from static-ct-api logs are permitted (Chrome 144+). ([Upcoming removal of Chrome's One-6962-log policy](#), as of 2025-12-01, confidence: high, source: grok)
- *New detail about removal of the log-diversity requirement, distinct from the earlier note that both log formats are admissible.*

Log incidents

- Tuscolo2026h2 static-CT log experienced 30-minute write-path degradation on August 9, 2026 due to traffic spike and lock starvation bug; all entries met 60s MMD. ([Tuscolo2026h2 write-path degradation on August 9th](#), as of 2026-08-10, confidence: high, source: grok)
- *New log incident report not previously recorded.*

Merkle Tree Certificates as the successor design

- IETF draft-ietf-plants-merkle-tree-certs reached version 05 (July 2026), describing Merkle Tree Certificates that integrate logging with issuance. ([draft-ietf-plants-merkle-tree-certs-05 - Merkle Tree Certificates](#), as of 2026-07-06, confidence: high, source: grok)
- *New version/status detail for the Merkle Tree Certificates successor design draft.*

The static-ct-api transition

- Chrome's policy change admitting static-ct-api logs took effect in Chrome 134, released 4 March, with log operators able to submit static-ct-api-compliant logs for inclusion from 1 April. ([groups.google.com — HAaVRK6MAAAJ](#), as of 2026-04-01, confidence: medium, source: ingest)
- *New timeline detail (Chrome 134, specific dates) for the static-ct-api rollout not previously recorded.*

Revocation mechanisms & timelines (~~revocation-mechanisms~~)

Newly captured background -- older material, first time in the knowledge base (14)

BR revocation timelines (24h / 5 day) and certificate problem report requirements

- SCWG discussions in 2025 proposed splitting TLS BR 4.9.1.1 Item 12 into separate triggers (BR noncompliance vs. CP/CPS deviations) and introducing a 90-day revocation timeline for the latter; related ballots (SC0XX Improve Certificate Problem Reports) remain in draft as of December 2025. ([Minutes of the F2F 65 Meeting in Toronto, Canada - SCWG, June 11, 2025](#), as of 2025-06-11, confidence: medium, source: grok)
- *New draft proposal details about splitting revocation triggers and a proposed 90-day timeline, not previously captured.*

CRLite and CRLSets

- Chrome CRLSets continue to cover only about 1% of all revocations (roughly 35k of 4m), while CRLite provides comprehensive coverage; no change to CRLSets scope reported through mid-2026. ([After OCSP: Certificate Revocation Done Right in 2026 — XICTRAQ Blog](#), as of 2026-07-22, confidence: medium, source: grok)
- *New quantitative comparison of CRLSets vs CRLite coverage not previously recorded in the knowledge base.*
- Firefox 142 began production use of CRLite in August 2025, delivering full Web PKI revocation coverage (99.84% of CT-visible certificates) via ~300 kB daily updates and local queries with no network lookup. ([CRLite: Fast, private, and comprehensive certificate revocation checking in Firefox - Mozilla Hacks](#), as of 2025-08-19, confidence: high, source: grok)
- *Adds specific Firefox version, production date, and coverage percentage not previously recorded.*

Current State

- CRLite encodes its revocation set in about 300 kB per day against roughly 600 kB for a recent CRLSet, so it delivers complete coverage in half the bandwidth. ([hacks.mozilla.org — crlite fast private and comprehensive certificate revocation checking in firefox](#), as of 2025-08-31, confidence: medium, source: ingest)
- *Introduces a new bandwidth-comparison detail between CRLite and CRLSets not previously recorded.*
- The efficiency difference is structural: CRLite uses about 6.6 bits per revocation against roughly 110 bits for CRLSet, because its filter cascade encodes membership rather than listing entries. ([hacks.mozilla.org — crlite fast private and comprehensive certificate revocation checking in firefox](#), as of 2025-08-31, confidence: medium, source: ingest)
- *Adds a new technical explanation for the efficiency gap not present in existing knowledge.*
- The two browsers solve different problems with these mechanisms. Mozilla pairs OneCRL for intermediate certificate revocations with CRLite for end-entity revocations, while Chrome's CRLSets are a selective list aimed at high-impact and emergency revocations rather than comprehensive coverage. ([hacks.mozilla.org — crlite fast private and comprehensive certificate revocation checking in firefox](#), as of 2025-08-31, confidence: medium, source: ingest)
- *Provides new architectural detail (OneCRL vs CRLite roles, CRLSets' selective-list purpose) absent from the existing knowledge base.*

Short-lived certificates as the replacement

- Let's Encrypt issued its first six-day certificate on 20 February 2025. ([letsencrypt.org — first short lived cert issued](#), as of 2025-02-20, confidence: medium, source: ingest)
- *New concrete milestone date for short-lived certificate issuance not previously recorded.*
- Six-day certificates carry neither OCSP nor CRL URLs. The compromise window is short enough that revocation is treated as unnecessary rather than as a service to be provided. ([letsencrypt.org — first short lived cert issued](#), as of 2025-02-20, confidence: medium, source: ingest)
- *New detail on how short-lived certificates substitute for revocation infrastructure, not previously captured.*

The end of OCSP at Let's Encrypt

- Let's Encrypt ended its OCSP service in August 2025, having already stopped including OCSP URLs in certificates more than 90 days earlier so that every certificate carrying one had expired before the responder went dark. ([letsencrypt.org — ocsp service has reached end of life](#), as of 2025-08-06, confidence: medium, source: ingest)
- *Adds the specific shutdown month and the URL-phaseout mechanism, beyond the baseline's generic 'ended OCSP in 2025'.*
- The stated reason for ending OCSP was privacy: the protocol tells the responder which site a user is visiting. ([letsencrypt.org — ocsp service has reached end of life](#), as of 2025-08-06, confidence: medium, source: ingest)
- *New rationale detail for the OCSP shutdown not present in baseline.*
- At its peak the service answered roughly 340 billion OCSP requests per month, with more than 140,000

requests per second handled by the CDN. That volume is the scale of the infrastructure the ecosystem has now removed. ([letsencrypt.org — ocsp service has reached end of life](#), as of 2025-08-06, confidence: medium, source: ingest)

- *New quantitative detail on OCSP traffic scale not previously recorded.*
- Let's Encrypt announced the intent to replace OCSP with CRLs in July 2024, more than a year before the shutdown, and now publishes revocation information exclusively by CRL. ([letsencrypt.org — replacing ocsp with crls](#), as of 2024-07-23, confidence: medium, source: ingest)
- *Adds the specific announcement date and confirms exclusive CRL publication, beyond the baseline's general 'moving to CRLs' statement.*

mass-revocation events and delayed-revocation incidents

- Mozilla Root Store Policy v3.0 requires CAs to develop, maintain, and annually test comprehensive mass revocation plans, with third-party assessment starting with June 2025 audits, and states there are no exceptions to TLS BR revocation timelines. ([CA/Responding To An Incident - MozillaWiki](#), as of 2025-09-01, confidence: high, source: grok)
- *New requirement detail about Mozilla's mass-revocation planning and audit mandate not previously recorded.*
- Mozilla updated its Root Store Policy in 2025 (v3.0) to require proactive subscriber warnings about revocation timelines and contractual language mandating timely cooperation, following multiple delayed-revocation incidents reported in Bugzilla. ([Enhancing CA Practices: Key Updates in Mozilla Root Store Policy, v3.0 - Mozilla Security Blog](#), as of 2025-03-12, confidence: high, source: grok)
- *New policy detail on subscriber warning and contractual cooperation requirements distinct from the mass-revocation-plan claim.*

Server Certificate Working Group

TLS BR ballots & versions (tls-br-ballots)

Superseded (4)

BR version releases on GitHub with redlines

- TLS Baseline Requirements version 2.2.9 was adopted via Ballot SC101v2 on 2026-07-02 and became effective on 2026-08-06. ([Latest Baseline Requirements | CA/Browser Forum](#), as of 2026-08-06, confidence: high, source: grok)
- supersedes: "TLS Baseline Requirements are at v2.2.8 (Ballot SC098, RFC 8657 CAA parameters, published 2026-06-16)"
- *The BR version has advanced from 2.2.8 to 2.2.9 following adoption of SC101v2, superseding the baseline's version number.*

SC-prefixed ballots through the full lifecycle

- Ballot SC101v2: Clarify Authorization Domain Names passed voting, completed IPR review, and was adopted effective 2026-08-06. ([Latest Baseline Requirements | CA/Browser Forum](#), as of 2026-08-06, confidence: high, source: grok)
- supersedes: "Later SC ballots (SC101v2 authorization domain names, SC102 EV domain reuse/validity alignment, SC103 EKUs for cross-certified sub-CAs, ML-DSA-87 keys) are proposals — VERIFY status"
- *SC101v2's status has advanced from proposal/VERIFY to fully adopted, resolving the baseline's flagged VERIFY status for this ballot.*
- Ballot SC102: EV Domain Reuse and Validity Alignment entered voting period as of early July 2026. ([Server Certificate WG \(CA/B Forum\) - Google Groups](#), as of 2026-07-07, confidence: medium, source: grok)
- supersedes: "Later SC ballots (...SC102 EV domain reuse/validity alignment...) are proposals — VERIFY"

status"

- *Gives a specific lifecycle stage (voting, as of July 2026) for SC102, replacing the vague 'proposal — VERIFY' status.*
- Ballot SC103: Require EKUs for Cross-Certified Subordinate CAs entered discussion period as of early July 2026. ([Server Certificate WG \(CA/B Forum\) - Google Groups](#), as of 2026-07-01, confidence: medium, source: grok)
- supersedes: "Later SC ballots (...SC103 EKUs for cross-certified sub-CAs...) are proposals — VERIFY status"
- *Gives a specific lifecycle stage (discussion, as of July 2026) for SC103, replacing the vague 'proposal — VERIFY' status.*

Newly captured background -- older material, first time in the knowledge base (4)

BR version releases on GitHub with redlines

- Redlined and clean versions of TLS BR v2.2.9 (from SC101v2) and v2.2.8 (from SC098) are maintained in the cabforum/servercert GitHub repository. ([Baseline Requirements for the Issuance and Management of Publicly-Trusted TLS Server Certificates](#), as of 2026-08-06, confidence: high, source: grok)
- *New detail identifying the specific GitHub repository location for BR redlines, not previously recorded.*

SC-prefixed ballots through the full lifecycle

- Ballot SC097: Sunset all remaining use of SHA-1 signatures in Certificates and CRLs was adopted 2026-02-24 with phased effective dates starting September 15, 2026. ([Ballot SC097: Sunset all remaining use of SHA-1 signatures in Certificates and CRLs | CA/Browser Forum](#), as of 2026-02-24, confidence: high, source: grok)
- *SC097 is not mentioned anywhere in the baseline or prior deltas; this is a wholly new ballot record.*
- Ballot SC095: Clean-up 2025 was adopted 2026-02-27 and effective 2026-03-31. ([Latest Baseline Requirements | CA/Browser Forum](#), as of 2026-03-31, confidence: high, source: grok)
- *SC095 is not mentioned anywhere in the baseline or prior deltas; this is a wholly new ballot record.*
- Ballot SC096: Carve-out for DNSSEC verification logging requirements was adopted 2026-01-14 and effective 2026-02-17. ([Latest Baseline Requirements | CA/Browser Forum](#), as of 2026-02-17, confidence: high, source: grok)
- *SC096 is not mentioned anywhere in the baseline or prior deltas; this is a wholly new ballot record.*

Validation Subcommittee & domain control (~~validation-subcommittee~~)

New developments (2)

Multi-Perspective Issuance Corroboration, in force and escalating

- The MPIC requirement escalates on a schedule rather than arriving at full strength: two corroborating perspectives from 15 September 2025, three from March 2026, four from June 2026, and five by December 2026. The December step is the next one due. ([sectigo.com — mpic faq](#), as of 2026-09-01, confidence: medium, source: ingest)
- *The escalation schedule for number of MPIC perspectives is not present anywhere in the baseline or prior deltas.*
- MPIC exists to defeat attacks that manipulate a single network path, including DNS response manipulation and BGP hijacking, by requiring the same validation result from multiple vantage points. ([github.com — 487](#), as of 2026-09-01, confidence: medium, source: ingest)
- *New rationale/threat-model detail for MPIC not previously recorded.*

Superseded (1)

Authorization domain names and validation method recording

- Ballot SC101 clarified Authorization Domain Names and produced Baseline Requirements version 2.2.9, effective 6 August 2026. This resolves the topic baseline's VERIFY marker on SC101v2: it is published, not a proposal. ([cabforum.org — requirements](#), as of 2026-08-06, confidence: medium, source: ingest)
- supersedes: "Later SC ballots (SC101v2 authorization domain names, SC102 EV domain reuse/validity alignment, SC103 EKUs for cross-certified sub-CAs, ML-DSA-87 keys) are proposals — VERIFY status."
- *Resolves the baseline VERIFY marker for SC101v2, confirming it is now adopted and published as BR v2.2.9.*

Newly captured background -- older material, first time in the knowledge base (10)

Authorization domain names and validation method recording

- Ballot SC099, adopted 18 April 2026 as version 2.2.7, improved the recording of which validation method was used for a given domain. ([cabforum.org — requirements](#), as of 2026-04-18, confidence: medium, source: ingest)
- *New ballot/version history detail on validation-method recording not present in baseline.*
- Ballot SC098, adopted 13 May 2026 as version 2.2.8, requires processing of RFC 8657 CAA parameters, which lets a domain holder restrict issuance to a named account or validation method. ([cabforum.org — requirements](#), as of 2026-05-13, confidence: medium, source: ingest)
- *Adds the ballot's adoption date (13 May 2026) and the substantive mechanism (account/method-restriction CAA parameters), which is distinct from and more detailed than the baseline's mere mention of a publication date.*

DNSSEC on CAA lookups

- Ballot SC100 consolidates DNSSEC validation rules that were scattered through the Baseline Requirements into a new Section 4.2.2.2, restructuring for readability and reducing repetition. ([cabforum.org — ballot sc100 dnssec clarification and consolidation](#), as of 2026-08-06, confidence: medium, source: ingest)
- *SC100 is not mentioned in baseline or prior deltas; this is a new ballot in progress.*
- SC100 clarifies that DNSSEC validation MAY be performed on Remote Network Perspectives but is REQUIRED only on the Primary Network Perspective. This is the interaction point between the DNSSEC requirement and MPIC. ([cabforum.org — ballot sc100 dnssec clarification and consolidation](#), as of 2026-08-06, confidence: medium, source: ingest)
- *Clarifies how the SC-085 DNSSEC-on-Primary-Perspective rule interacts with MPIC's multiple perspectives — a new detail beyond baseline's SC-085 statement.*
- SC100 is adopted and in IPR review from 6 August to 5 September 2026, so the consolidated Section 4.2.2.2 is not yet published text. ([cabforum.org — ballot sc100 dnssec clarification and consolidation](#), as of 2026-08-06, confidence: medium, source: ingest)
- *New status information tracking SC100 through the ballot lifecycle, not previously recorded.*
- Version 2.2.4 of the Baseline Requirements, adopted by Ballot SC096 on 14 January 2026, created a carve-out for DNSSEC verification logging. ([cabforum.org — requirements](#), as of 2026-01-14, confidence: medium, source: ingest)
- *Fills in a prior BR version/ballot in the version history not present in baseline, which only starts detailed tracking at v2.2.8.*
- Version 2.2.3, adopted by Ballot SC094 on 15 January 2026, created a DNSSEC exception in the email domain control validation methods. ([cabforum.org — requirements](#), as of 2026-01-15, confidence: medium, source: ingest)
- *New ballot/version history detail not present in baseline.*

Method sunsets and reuse periods

- Ballot SC090, adopted 20 November 2025 as version 2.2.2, gradually sunsets the email-based and phone-

based domain validation methods. ([cabforum.org — requirements](#), as of 2025-11-20, confidence: medium, source: ingest)

- *New information on the sunset of email/phone-based DV methods, not present in baseline.*
- Ballot SC102 removes the WHOIS and RDAP same-registrant check from EV domain revalidation, and replaces hardcoded 398-day reuse values in the EV Guidelines with references to the Baseline Requirements, so that EV reuse now tracks the BR schedule automatically. ([cabforum.org — ballot sc102 ev domain reuse and validity alignment](#), as of 2026-07-14, confidence: medium, source: ingest)
- *Baseline only names SC102 as a VERIFY proposal without describing content; this adds substantive detail on what the ballot changes, without asserting adoption status.*

Multi-Perspective Issuance Corroboration, in force and escalating

- Multi-Perspective Issuance Corroboration became mandatory for all publicly trusted TLS certificates on 15 September 2025 under Ballot SC-067, requiring domain validation and CAA checks to be corroborated from geographically and topologically distinct network perspectives. ([github.com — 487](#), as of 2025-09-15, confidence: medium, source: ingest)
- *Baseline only states SC-067/MPIC is 'in force' with no date or mechanism detail; this supplies the specific effective date and what is corroborated, a genuinely new detail.*

Certificate profiles & cryptographic requirements (cert-profiles-crypto)

New developments (2)

Post-quantum keys: proposed, not balloted

- A draft ballot to permit ML-DSA public keys and signatures in publicly trusted certificates exists as a pull request against the cabforum/servercert repository, covering ML-DSA-44, ML-DSA-65 and ML-DSA-87 parameter sets; it is draft text, not a ballot in any lifecycle stage. ([github.com — 662](#), as of 2026-09-01, confidence: medium, source: ingest)
- *New, more detailed information about pre-ballot ML-DSA drafting activity not previously recorded.*

Profile changes in flight

- Ballot SC103, which would require Extended Key Usage extensions on cross-certified Subordinate CAs, remains in discussion stage with unresolved questions on the mailing list (not yet advanced to voting). ([cabforum.org — ballots](#), as of 2026-09-01, confidence: medium, source: ingest)
- *Clarifies SC103's concrete lifecycle stage (discussion, unresolved) beyond the baseline's generic 'proposal — VERIFY status'.*

Newly captured background -- older material, first time in the knowledge base (3)

Algorithm and signature requirements

- Ballot SC097, adopted 24 February 2026 as Baseline Requirements version 2.2.5, sunsets all remaining use of SHA-1 signatures. ([cabforum.org — requirements](#), as of 2026-02-24, confidence: medium, source: ingest)
- *New fact about a SHA-1 sunset ballot and its adoption date/version not present in baseline or prior claims.*

Post-quantum keys: proposed, not balloted

- As of May 2026, no merged Baseline Requirement permits ML-DSA in publicly trusted certificates, and no public root CA has issued an ML-DSA root chaining into the Mozilla, Apple, Microsoft or Chrome trust stores. ([cabforum.org — 2026 05 21 minutes of the server certificate working group](#), as of 2026-05-21, confidence: medium, source: ingest)

- Adds the specific, checkable fact that no BR currently permits ML-DSA (beyond the baseline's more general no-PQ-root statement).
- Post-quantum certificate authentication has two blocking dependencies, only one of which sits with the Forum: the Baseline Requirements must be amended to permit ML-DSA, and the IETF must finalize OID and encoding standards for post-quantum algorithms in X.509. ([cabforum.org — 2026 05 21 minutes of the server certificate working group](#), as of 2026-05-21, confidence: medium, source: ingest)
- New explanatory detail on why ML-DSA adoption is blocked, distinguishing Forum-side from IETF-side dependencies, not previously captured.

Code Signing Certificate Working Group

Code Signing BR ballots & versions (csbr-ballots)

Superseded (2)

CSBR versions

- Code Signing Baseline Requirements updated to version 3.11.0 incorporating Ballot CSC-32 (Mandatory inclusion of a Reserved Policy OID for Subscriber certificates), with effective date 15 September 2026. ([Latest Code Signing Baseline Requirements](#), as of 2026-09-01, confidence: high, source: grok)
- supersedes: "Code Signing Baseline Requirements are at v3.8 (published 2024-08-05, EV Code Signing Guidelines merged in)"
- Confirms the CSBR version has moved from the baseline's v3.8 to v3.11.0, publishing the previously-passed CSC-32 reserved-OID requirement with a concrete effective date.

Passed ballots not yet in a published version

- Ballot CSC-32, "Make a Reserved Policy OID mandatory", passed on 16 June 2026, resolving the prior VERIFY status. ([cabforum.org — ballots](#), as of 2026-06-16, confidence: medium, source: ingest)
- supersedes: "CSC-32 (mandatory reserved policy OID) is a proposal — VERIFY"
- Confirms CSC-32 passed voting, resolving the baseline's VERIFY marker on its status.

Newly captured background -- older material, first time in the knowledge base (3)

Failed ballots

- Ballot CSC-030, the June 2025 Server Certificate Working Group alignment ballot, failed. ([cabforum.org — ballots](#), as of 2025-06-30, confidence: medium, source: ingest)
- New information about a failed alignment ballot not previously in the knowledge base.
- Ballot CSC-029, the March 2025 Server Certificate Working Group alignment ballot, also failed, marking two consecutive failed alignment ballots. ([cabforum.org — ballots](#), as of 2025-03-31, confidence: medium, source: ingest)
- New information establishing a pattern of failed SCWG-alignment ballots in the Code Signing group.

Passed ballots not yet in a published version

- Ballot CSC-31, "Maximum Validity Reduction", passed on 17 November 2025. ([cabforum.org — ballots](#), as of 2025-11-17, confidence: medium, source: ingest)
- Not previously recorded in the empty leaf snapshot; a passed ballot on code signing cert validity reduction is new information.

S/MIME Certificate Working Group

S/MIME BR ballots & versions (**smime-br-ballots**)

New developments (1)

Current version and the RSA key size transition

- From 15 September 2027, CAs shall not issue subscriber certificates from any Subordinate CA whose RSA modulus is less than 3072 bits; this sunsets issuance from legacy 2048-bit Sub-CAs without requiring their immediate revocation. ([cabforum.org — requirements](#), as of 2027-09-15, confidence: medium, source: ingest)
- *Adds a new operational detail beyond the baseline date/threshold: legacy sub-CAs need not be revoked immediately, only barred from new issuance.*

Superseded (1)

Adopted, in IPR review

- Ballot SMC018, "Realignment of Multipurpose use cases," was posted on 11 August 2026 and is in its 30-day IPR review period, having passed voting. ([cabforum.org — smime](#), as of 2026-08-11, confidence: medium, source: ingest)
- supersedes: "SMC018 (cross-cert and ECU restrictions) was in discussion — VERIFY"
- *Updates SMC018's lifecycle status from 'in discussion' (baseline VERIFY marker) to having passed voting and entering the 30-day IPR review period.*

Newly captured background -- older material, first time in the knowledge base (1)

Current version and the RSA key size transition

- Version 1.0.14 of the S/MIME Baseline Requirements was published in May 2026 by Ballot SMC016. ([cabforum.org — requirements](#), as of 2026-05-05, confidence: medium, source: ingest)
- *Introduces a prior SBR version and ballot (SMC016/v1.0.14) not present anywhere in baseline or prior deltas.*